

Array Diagnostic test

You must show all your working in the following. No correct table (exactly the same as the solution)... no marks. If you do not get them all right and do not understand why, ask for help.

1. What is printed when the following code fragment has executed?

```
int[]age = new int[4];           // L1
System.out.println(age[0]);      // L2
age[0] = 18;                     // L3
age[1] = 19;                     // L4
age[2] = 21;                     // L5
age[3] = 17;                     // L6
System.out.println(age[0]);      // L7
```

Printed:

0
18

LINE	age				Printed
	[0]	[1]	[2]	[3]	
L1	0	0	0	0	
L2					0
L3	18				
L4		19			
L5			21		
L6				17	
L7					18

2. What is printed when the following code fragment has executed?

```
int[]age = new int[2];           // L1
age[0] = 33;                     // L2
age[1] = 27;                     // L3
age[1] = 28;                     // L4
age[0] = 42;                     // L5
System.out.println(age[1]);      // L6
```

Printed:

28

LINE	age		Printed
	[0]	[1]	
L1	0	0	
L2	33		
L3		27	
L4		28	
L5	42		
L6			28

3. What is printed when the following code fragment has executed?

```
int[]age1 = new int[2];          // L1
int[]age2 = new int[2];          // L2
age1[0] = 60;                     // L3
age1[1] = 65;                     // L4
age2[1] = age1[0];                // L5
age2[0] = age1[1];                // L6
System.out.println(age2[1]);      // L7
```

Printed:

60

LINE	age1		age2		Printed
	[0]	[1]	[0]	[1]	
L1	0	0			
L2			0	0	
L3	60				
L4		65			
L5				60	
L6			65		
L7					60

4. What is printed when the following code fragment has executed?

```
int x = 5; // L1
int[] s = new int[2]; // L2

for (int i = 0; i < 2; i++) // L3
{
    System.out.println(s[i]); // L4
    s[i] = s[i] + x; // L5
}
System.out.println(s[0]); // L6
```

Printed:

0
0
5

LINE	x	s		i	i < 2?	Printed
		[0]	[1]			
L1	5					
L2		0	0			
L3				0	0 < 2? true	
L4						0
L5		5				
L3				1	1 < 2? true	
L4						0
L5			5			
L3					2 < 2? false	
L6						5

5. What is printed when the following code fragment has executed?

```
int[] s = new int[3]; // L1

for (int i = 2; i > 0; i--) // L2
{
    s[i] = i+4; // L3
    System.out.println(s[i]); // L4
}
```

Printed:

6
5

LINE	s			i	i > 0?	Printed
	[0]	[1]	[2]			
L1	0	0	0			
L2				2	2 > 0? true	
L3			6			
L4						6
L2				1	1 > 0? true	
L3		5				
L4						5
L2				0	0 > 0? false	

6. What is printed when the following code fragment has executed?

```
double[] shoe = new double[1]; // L1
System.out.println(shoe[0]); // L2
shoe[0] = 9.5; // L3
System.out.println(shoe[0]); // L4
```

Printed:

0.0
9.5

LINE	shoe	Printed
	[0]	
L1	0.0	
L2		0.0
L3	9.5	
L4		9.5